JAVA • MYSQL • JDBC

PUBLIC CANTEEN MANAGEMENT SYSTEM

A command-line application for managing food, customers, orders, billing and canteen records.

PROJECT PRESENTATION

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Problem statement & project objectives

The challenge

Traditional canteen records may be maintained manually.

- Food and stock details take time to update
- Billing can involve calculation errors
- Previous orders are difficult to retrieve
- Paper-based records are harder to organise

What the system aims to do

Provide a computerised way to manage canteen operations.

- Manage food items with CRUD operations
- Store customer and order information
- Calculate bills automatically
- Update stock after successful orders
- Maintain order history and basic sales reports

PROJECT GOAL **Reduce manual work and keep canteen information organised.**

System overview & key features

The application runs through Windows Command Prompt. Java handles application logic, MySQL stores records, and JDBC connects the two.

Administrator login
01

Checks credentials before showing the main menu.

Food management
02

Add, view, search, update and delete food records.

Customer details
03

Store customer name and phone information.

Orders & billing
04

Record selected items and quantities; calculate totals.

Stock management
05

Reduce available quantity after a successful order.

History & reports
06

Retrieve previous orders and basic sales information.

Requirements & feasibility

Hardware

- Intel or equivalent processor
- 4 GB RAM or above recommended
- Sufficient storage, monitor, keyboard and mouse

Software

- Windows operating system
- Java JDK and text editor / IDE
- MySQL database and JDBC driver
- Command Prompt

FEASIBILITY

Technical — Java, MySQL and JDBC suit a database-driven application.

Operational — A menu-driven CMD interface can be used by an administrator.

Economic — Uses commonly available software without expensive infrastructure.

Schedule — Can be developed in stages: analysis, design, coding, testing and documentation.

System architecture

A simple application-to-database architecture connects the administrator interface to MySQL.



Request / response flow

Administrator input → Java processing → JDBC query → MySQL operation → result returned and displayed in CMD.

Main modules & application workflow

Login

Validate administrator credentials

Food

Maintain ID, name, price and quantity

Customer

Store name and phone number

Order

Select food items and quantities

Billing

Calculate subtotals and total

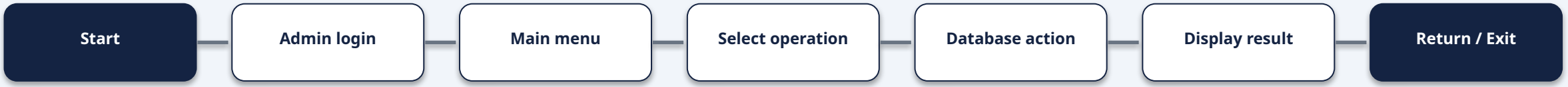
Stock

Update available quantity

Reports

View order history and sales totals

WORKFLOW



Database design

CanteenDB contains five main tables connected through primary and foreign keys.

ADMIN

Admin_ID (PK)
Username • Password

FOOD

Food_ID (PK)
Food_Name • Price • Quantity

CUSTOMER

Customer_ID (PK)
Customer_Name • Phone

ORDERS

Order_ID (PK)
Customer_ID (FK)
Order_Date • Total • Payment_Mode

1 customer → many orders

Food items can appear in many order-detail records.

ORDER_DETAILS

Detail_ID (PK)
Order_ID (FK) • Food_ID (FK)
Quantity • Subtotal

Normalisation separates entities into tables to reduce duplication and improve consistency (1NF, 2NF and 3NF principles).

Keys link CUSTOMER → ORDERS → ORDER_DETAILS, and FOOD → ORDER_DETAILS.

Implementation: Java, JDBC & billing

Java

• Implements application logic
• Provides menu-driven operations
• Uses classes, objects and methods to organise code

JDBC

• Java Database Connectivity API
• Connects Java with MySQL
• Executes SQL statements and retrieves or modifies records

MySQL

• Stores food and customer data
• Stores orders and order details
• Uses keys to link related records

BILLING LOGIC **Item subtotal = Food price × Ordered quantity | Total bill = Sum of all item subtotals**

Stock quantity is reduced after a successful order.

Testing, security & limitations

Verification plan

Test cases include:

• Valid and invalid admin login
• Add, view, search, update and delete food
• Place order and calculate bill
• View order history
• Generate sales report

Compare actual output with expected results.

Security & maintenance

• Validate user input
• Use PreparedStatement for SQL user input
• Keep database credentials private
• Hash passwords in real-world use
• Back up database regularly
• Maintain compatible Java and MySQL versions

Current limitations

• Command Prompt interface; no GUI
• Intended mainly for small canteens
• No online ordering or payment gateway
• No customer mobile application
• No advanced inventory management

Testing note: actual test outcomes should be confirmed by running the project; the report lists expected results.

Conclusion & future enhancements

Conclusion

The project demonstrates how Java, MySQL and JDBC can be combined to organise canteen operations.

It supports food management, customer records, order processing, billing, stock updates, order history and basic sales reporting.

Possible future work

- Graphical user interface
- Online food ordering and payment
- Customer login and registration
- Receipt printing and low-stock alerts
- Supplier and multi-branch support
- Web or mobile application

THANK YOU

Questions?

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